

Evaluation of a liquid media MALDI-TOF MS protocol for the identification of dermatophytes isolated from *tinea capitis* infections

P. Lecerf¹ • R. De Paepe² • Y. Jazaeri¹ • A.C. Normand³ • A. Packeu^{2, 4}

1. Faculty of Medicine, Université Libre de Bruxelles, Brussels, Belgium • 2. Service of Mycology and Aerobiology, Sciensano, Brussels, Belgium • 3. Service of Parasitology/Mycology, AP-HP Hôpitaux de Paris, Paris, France • 4. BCCM/IHEM Fungal Collection, Sciensano, Brussels, Belgium

CONCLUSION

A liquid media MALDI-TOF MS protocol was developed with the aim to create a quick and user friendly method for the identification of dermatophyte isolates from *tinea capitis* infections. When comparing the method to the standard MALDI TOF MS protocol described by Cassagne et al. (2011) based on solid cultures, the rate of correct identification (RCI) of the isolates was higher when using Cassagne’s protocol. This was the case for three different fungal databases. There was no significant difference in turn-around time of both methods, even when eliminating the sub-culturing of the isolates. An up-to-date database and a standardized protocol for both (clinical) samples and database reference strains remain vital in the use of MALDI-TOF MS as an identification tool for dermatophytes.

Tinea capitis is a common childhood superficial infection of the scalp caused by dermatophytes. Recently, Matrix-Assisted Laser Desorption Ionization Time-Of-Flight Mass Spectrometry (MALDI-TOF MS) has become a powerful tool in the rapid identification of these micro-organisms. Although MALDI-TOF MS provides a faster identification of dermatophytes than conventional methods based on morphological characteristics, identification can still take a couple of weeks due to slow growing cultures or sub-culturing steps. In this study, a liquid media MALDI-TOF MS protocol was developed to obtain a straightforward and user friendly identification protocol for clinical isolates of dermatophytes with a shorter turn-around time than the protocol based on solid cultures described by Cassagne et al. (2011).

Methods

- A MALDI-TOF MS protocol for the identification of dermatophytes causing *tinea capitis* was developed, using liquid cultures of the dermatophyte isolates.
- A comparison in rate of correct identification (RCI, correctly identified percentage of the samples) and turn-around time with the MALDI-TOF MS protein extraction protocol described by Cassagne et al. (2011) was made.
- The liquid MALDI-TOF MS protocol was validated by using it on reference strains of 10 common causal agents of *tinea capitis* from the BCCM fungal culture collection.
- The protocol was then tested on clinical isolates from suspected *tinea capitis* infections, while culturing these isolates with ($n = 39$) and without ($n = 22$) a sub-culturing step to evaluate if this step can be eliminated.
- Three databases were used to identify the detected spectra: the in-house BCCM/IHEM database, the Bruker filamentous fungi database 3.0 and the MSI V2.0 database.
- Blank samples of the liquid medium were analysed to determine if the medium interferes with the detection of spectra.
- The recorded spectra were visualized and compared to each other using the FlexAnalysis 3.4 software by Bruker Daltoniks.

Results

Reference strains of *Epidermophyton floccosum*, *Microsporum audouinii*, *Microsporum canis*, *Nannizzia gypsea*, *Trichophyton interdigitale*, *Trichophyton mentagrophytes*, *Trichophyton rubrum*, *Trichophyton soudanense*, *Trichophyton tonsurans* and *Trichophyton violaceum* were used to validate the liquid media MALDI-TOF MS protocol. Nine out of ten strains were correctly identified on species-level with the developed liquid protocol, but *Trichophyton tonsurans* was consistently misidentified as *Trichophyton interdigitale*.

The detected protein spectra of the liquid culture of *T. tonsurans* were compared to the reference spectra of this species from the in-house database (based on solid cultures) (Figure 1A). When compared to the reference spectra of *T. interdigitale* (Figure 1B), it can be seen that the peaks of the curves were indeed more concurring than in Figure 1A. The detected spectra originating from the liquid cultures showed clear differences with those originating from solid cultures. Analysis of blank samples of the medium indicate that the liquid medium interferes with the signal of the fungus, although the fungal pellet was washed several times.

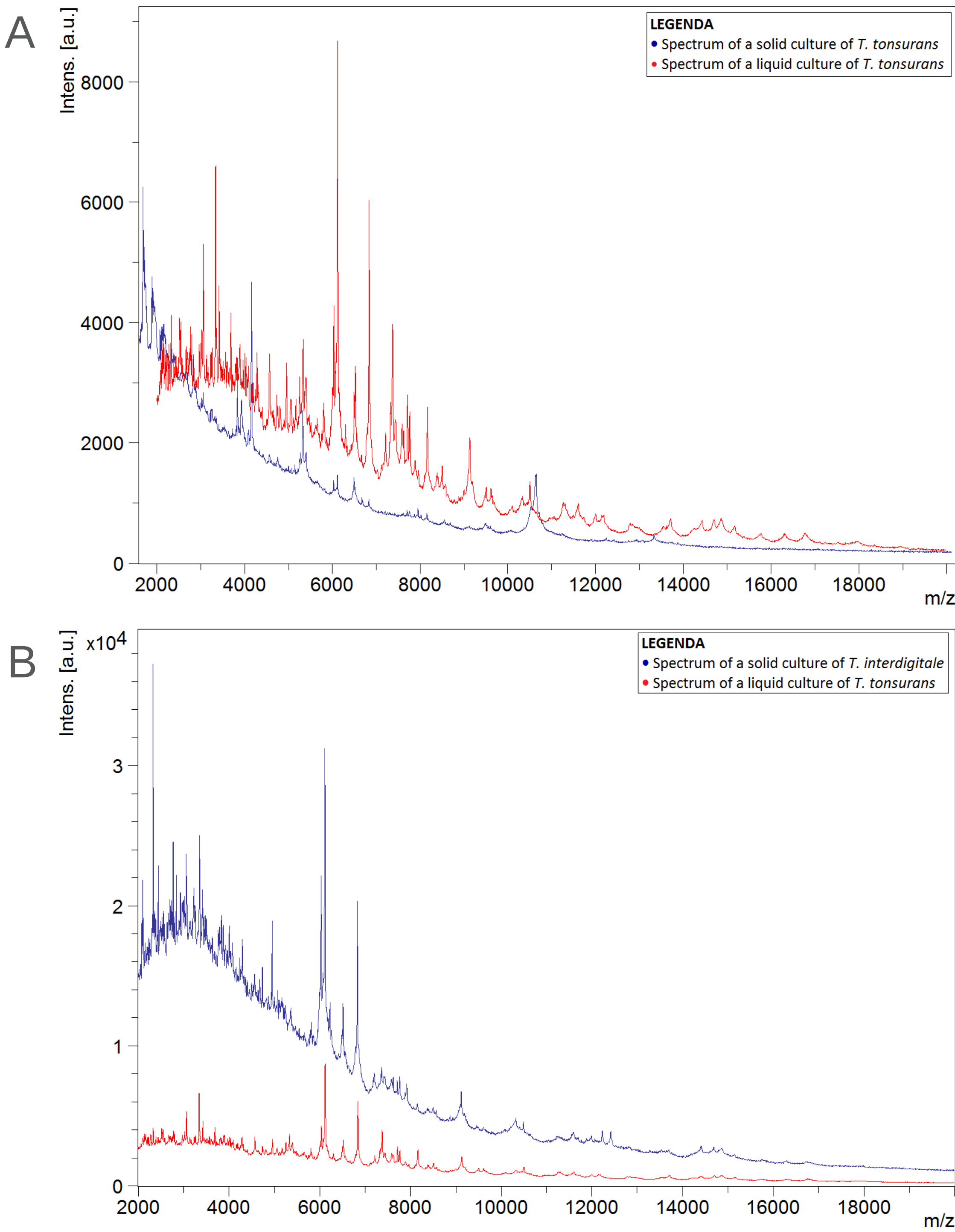


Figure 1: Protein spectrum of a liquid culture of a *Trichophyton tonsurans* reference strain (red curve in both graphs) compared to the protein spectra of solid reference cultures of *Trichophyton tonsurans* (A) and *Trichophyton interdigitale* (B) (blue curves). Protein spectra were visualized by plotting the signal intensity (in absorption units) of the ions in function of their mass-to-charge ratio (m/z, in kg/C).

When comparing the RCI of the liquid media MALDI-TOF MS protocol with the standard protocol by Cassagne et al., it could be noticed that the standard protocol led to a higher number of correct identifications, whichever database was applied (Table 1), with the exception of sub-cultured isolates in case of the Bruker database. Moreover, by skipping the sub-culturing step in the liquid protocol, the RCI became even lower for all databases. The turn-around time for identification of primary isolates was not significantly lower than for identification of sub-cultured isolates (respectively **12** and **14** days). Except for the liquid cultures of primary isolates, the RCI obtained with the in-house database was the highest, highlighting the importance of an up-to-date database with reference spectra created in the same manner as the spectra of sample isolates.

	Sub-cultures of clinical dermatophyte isolates		Primary dermatophyte isolates	
	RCI MALDI LC (%)	RCI MALDI SC (%)	RCI MALDI LC (%)	RCI MALDI SC (%)
In-house BCCM/IHEM database	62	87	41	94
MSI V2.0 database	47	68	45	61
Bruker filamentous fungi database	50	47	5	11

Table 1: Rate of correct identification (RCI, in %) of solid (SC) and liquid (LC) cultures of sub-cultured and primary clinical dermatophyte isolates. Tested for three different fungal databases.

REFERENCES

- Cassagne, C. et al. (2011). Mould routine identification in the clinical laboratory by matrix-assisted laser desorption ionization time-of-flight mass spectrometry. *PLoS One*, 6(12), e28425.

